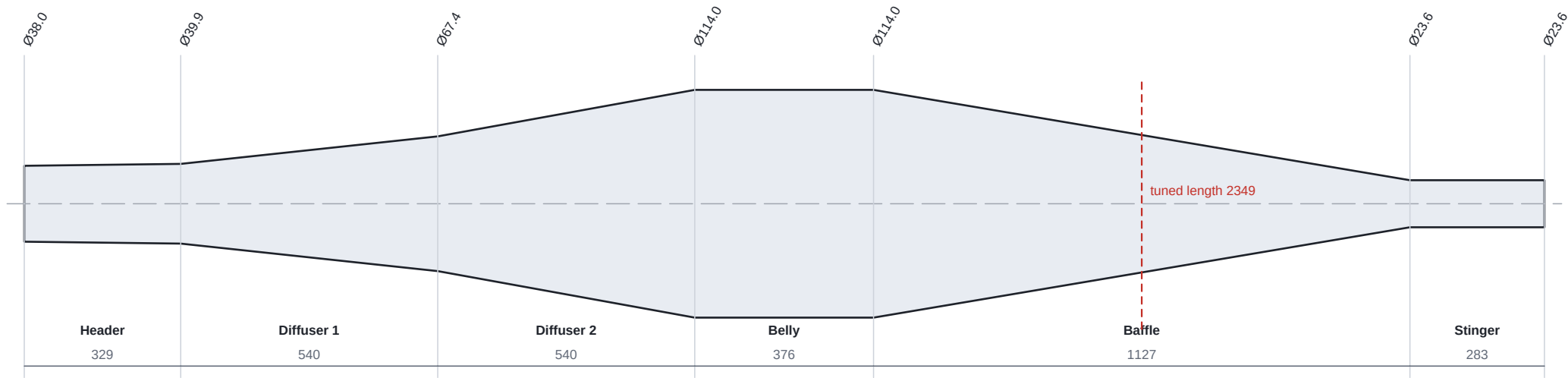


JLO L372 generator chamber

JLO L372 · 372 cc, 1 cyl · Power at a fixed rpm at 2850 rpm · Rolled and welded, 1.0 mm Mild steel



Sections

Section	Start	Length	Ø in	Ø out	Half-angle
Header	0	328.8	38.0	39.9	0.17°
Diffuser 1	329	540.2	39.9	67.4	1.46°
Diffuser 2	869	540.2	67.4	114.0	2.47°
Belly	1409	375.8	114.0	114.0	0.00°
Baffle	1785	1127.4	114.0	23.6	2.30°
Stinger	2912	282.7	23.6	23.6	0.00°
Overall		3195.1			

Key figures

Design speed	2850 rpm
Pipe tuned for	2850 rpm
Plugging pulse on time	2666 – 3034 rpm
Tuned length (port to mid-baffle)	2348.7 mm
Wave speed	518 m/s at 420 °C
Exhaust opens / closes	102° / 258° ATDC (155°)
Outlet / belly / stinger Ø	38.0 / 114.0 / 23.6 mm
Chamber volume	13.96 L
Sheet parts	0.67 m², 5.2 kg
Sheets	1 × 2500 × 1250

Check

The chamber is 3.20 m long, which is normal at 2850 rpm. Cut the cones into segments so the chamber can be bent to fit the frame.

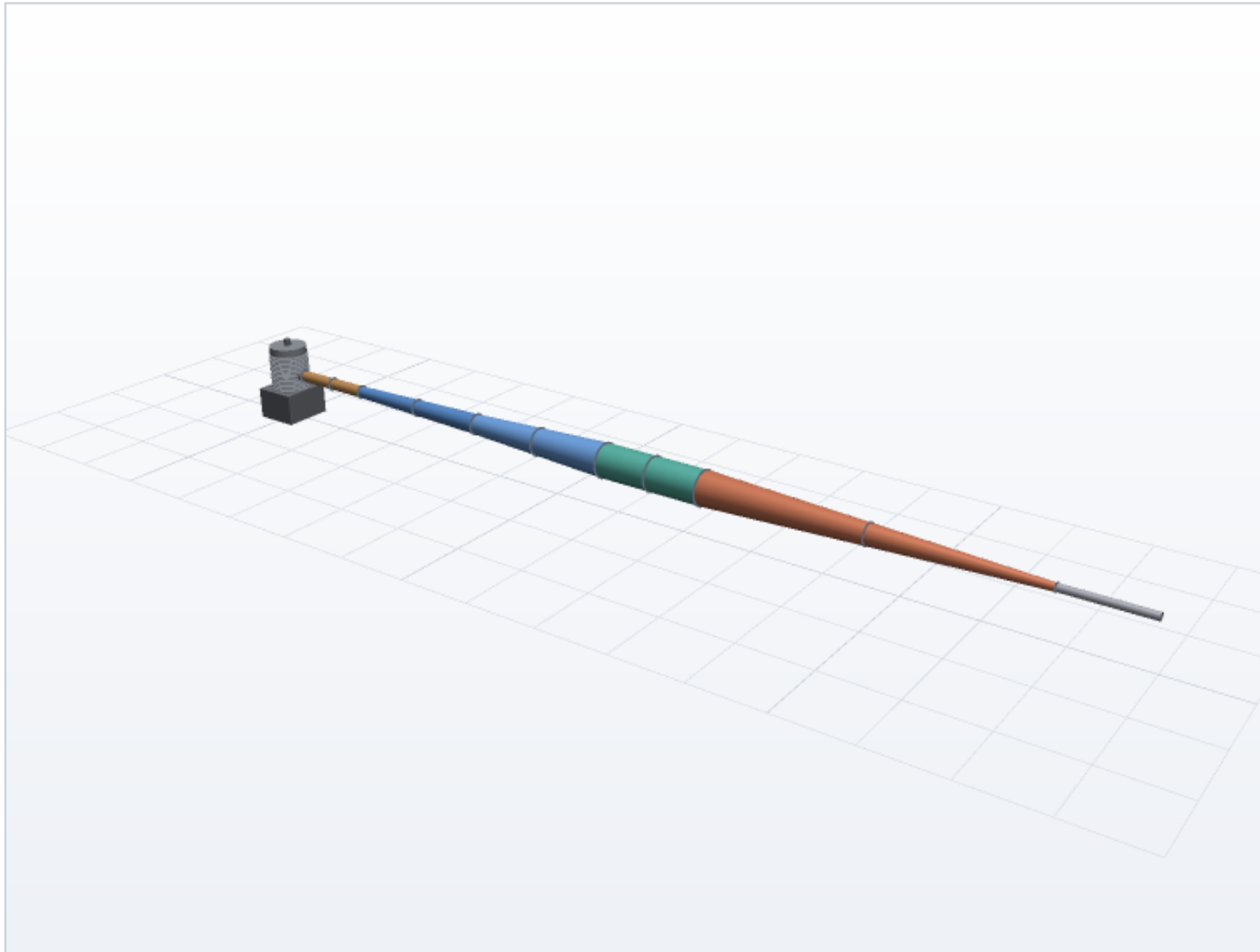
port face at left · vertical scale ×4.2

A starting-point design from wave-timing rules (Jennings/Blair tradition), not a gas-dynamic simulation. Confirm exhaust timing and outlet bore on the engine itself, and expect to fine-tune header and stinger length on the dyno.

JLO L372 generator chamber

3D VIEW AND ROUTE

JLO L372 · 372 cc, 1 cyl · Power at a fixed rpm at 2850 rpm · Rolled and welded, 1.0 mm Mild steel



Envelope

Chamber X × Y × Z
Along the centreline

3195 × 116 × 116 mm

3195 mm

No clashes

Bends

Straight. Bend it to fit on the Design page.

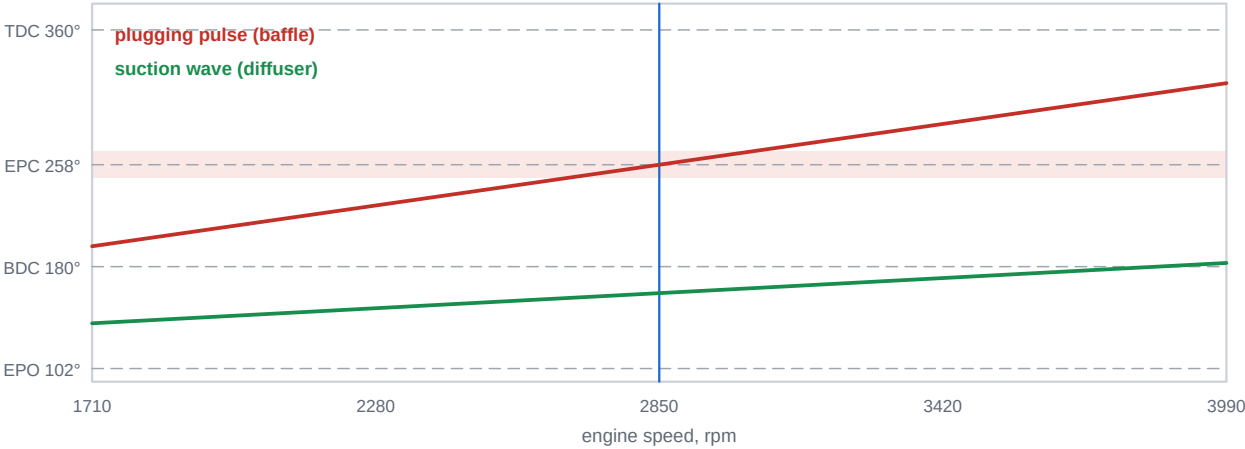
Engine coordinates: +X out of the exhaust port, +Y up the cylinder, Z along the crankshaft. The engine is drawn for scale only.

JLO L372 generator chamber

WAVE TIMING AND CUT LIST

JLO L372 · 372 cc, 1 cyl · Power at a fixed rpm at 2850 rpm · Rolled and welded, 1.0 mm Mild steel

When the reflections reach the port



Each curve is the crank angle at which a wave launched as the exhaust port opens gets back to it, reflected from the middle of the diffuser (suction) or of the baffle (plugging), at the speed along the bottom. The shaded band is 10° either side of exhaust closure: the plugging pulse crosses it between 2666 and 3034 rpm. The blue line is the design speed. Suction should arrive while the transfers are open, around BDC.

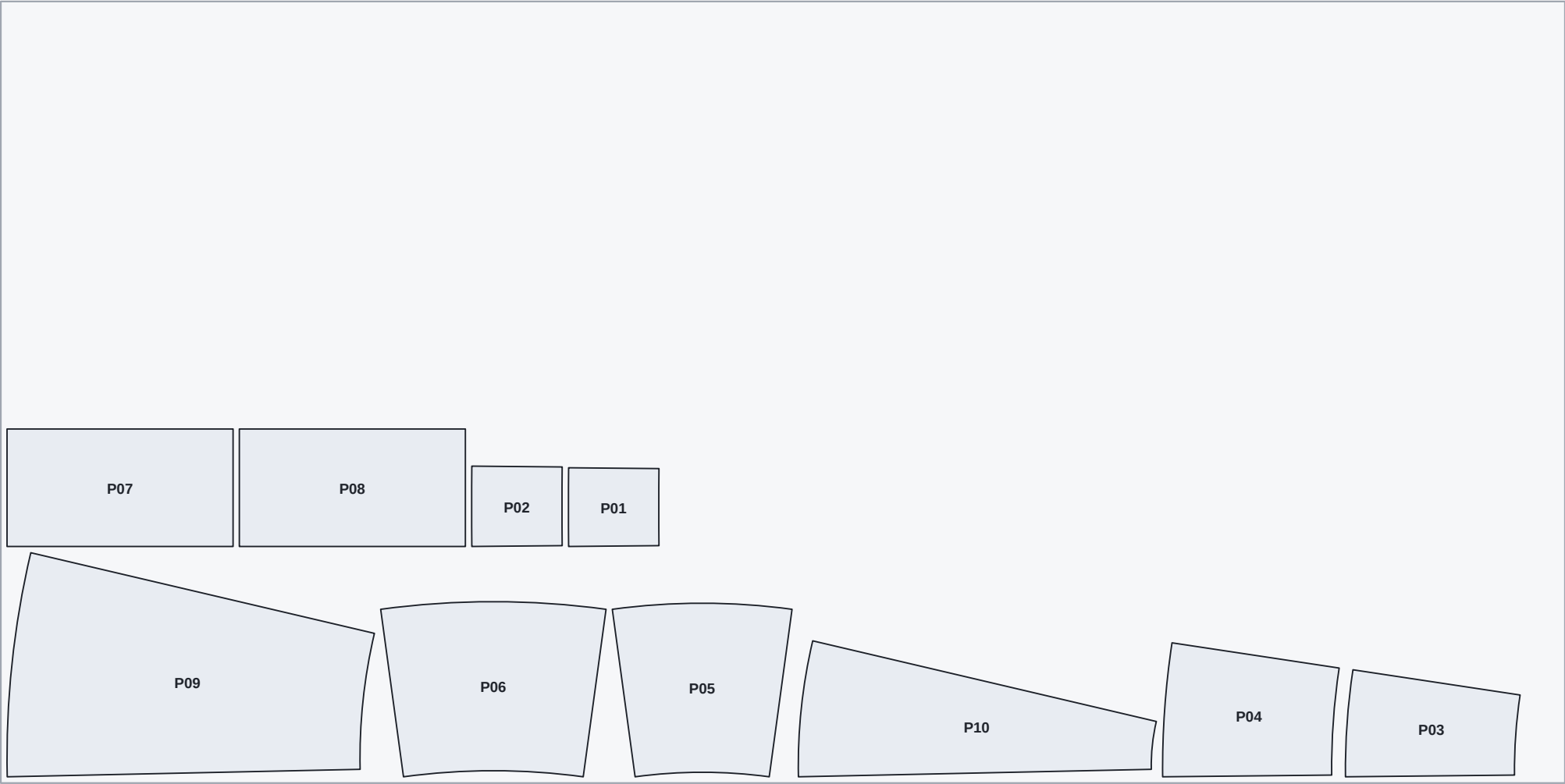
Cut list

ID	Part	Kind	Qty	Rolled Ø (mean)	Axial	Flat	Area m²
P01	Header (1 of 2)	cone	1	39.2 to 40.1	144.4	R6933.9 / R6789.4, 1.04°	0.018
P02	Header (2 of 2)	cone	1	40.1 to 40.9	144.4	R7078.3 / R6933.9, 1.04°	0.018
P03	Diffuser 1 (1 of 2)	cone	1	40.9 to 54.7	270.1	R1072.6 / R802.4, 9.17°	0.041
P04	Diffuser 1 (2 of 2)	cone	1	54.7 to 68.4	270.1	R1342.8 / R1072.6, 9.17°	0.052
P05	Diffuser 2 (1 of 2)	cone	1	68.4 to 91.7	270.1	R1065.2 / R794.9, 15.50°	0.068
P06	Diffuser 2 (2 of 2)	cone	1	91.7 to 115.0	270.1	R1335.6 / R1065.2, 15.50°	0.088
P07	Belly (1 of 2)	cylinder	1	115.0	187.9	361.3 × 187.9	0.068
P08	Belly (2 of 2)	cylinder	1	115.0	187.9	361.3 × 187.9	0.068
P09	Baffle (1 of 2)	cone	1	69.8 to 115.0	563.7	R1434.7 / R870.5, 14.43°	0.164
P10	Baffle (2 of 2)	cone	1	24.6 to 69.8	563.7	R870.5 / R306.4, 14.43°	0.084
P11	Stinger tube	tube	1 tube 25.6 OD × 1.0 wall		282.7	cut to length	—

JLO L372 generator chamber

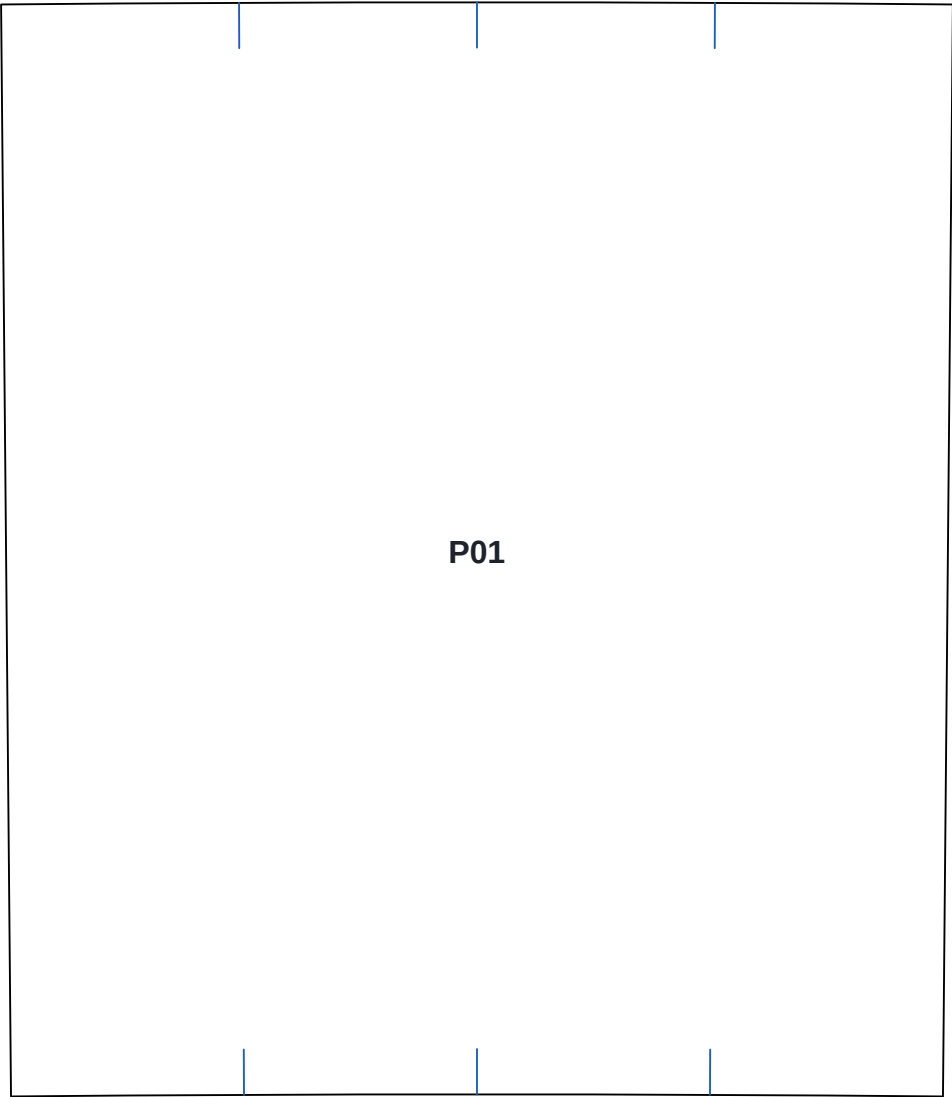
JLO L372 · 372 cc, 1 cyl · Power at a fixed rpm at 2850 rpm · Rolled and welded, 1.0 mm Mild steel

2500 × 1250 mm sheet, 1.0 mm Mild steel, drawn at 1:9



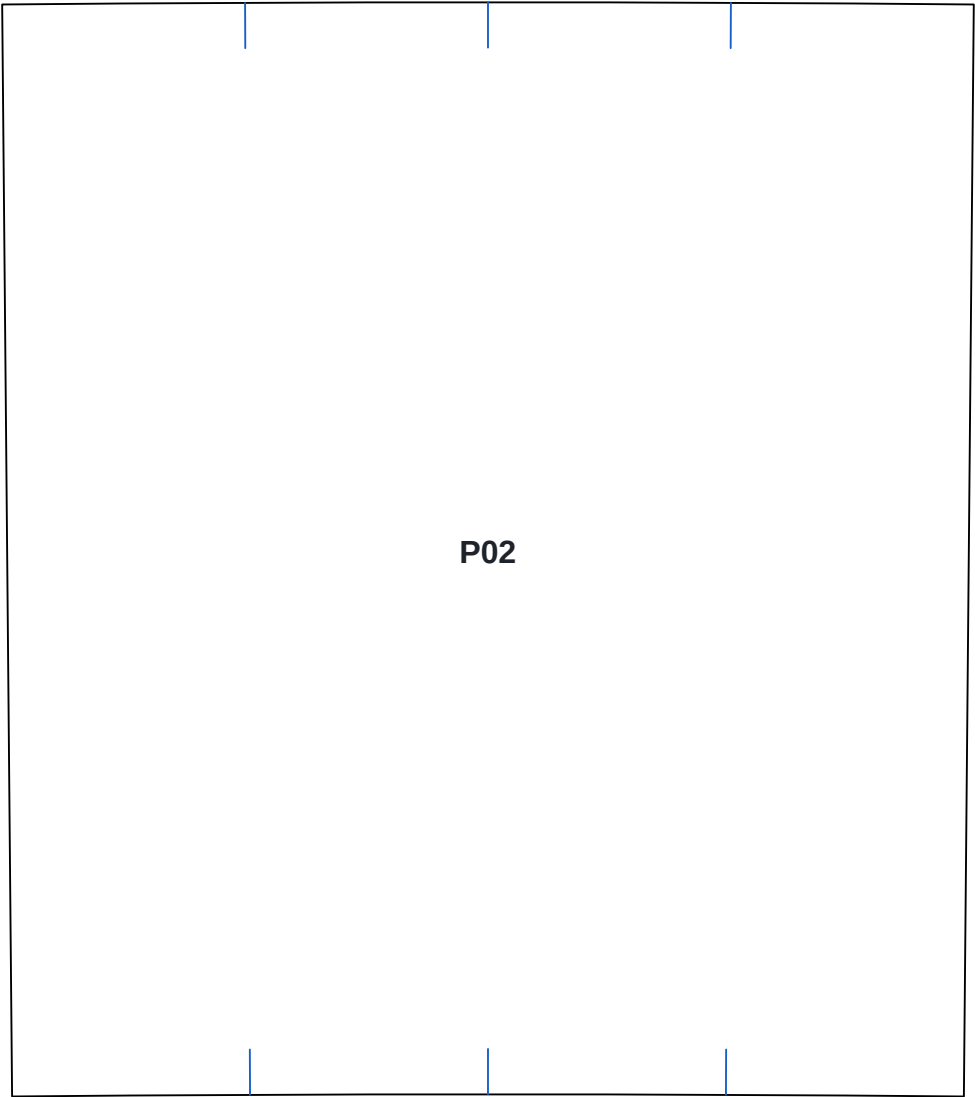
P01 Header (1 of 2)

Qty 1 · cone Ø39.2 to Ø40.1 mean, 144.4 long · outer R6933.85, inner R6789.44, 1.040° sector, slant 144.41
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14



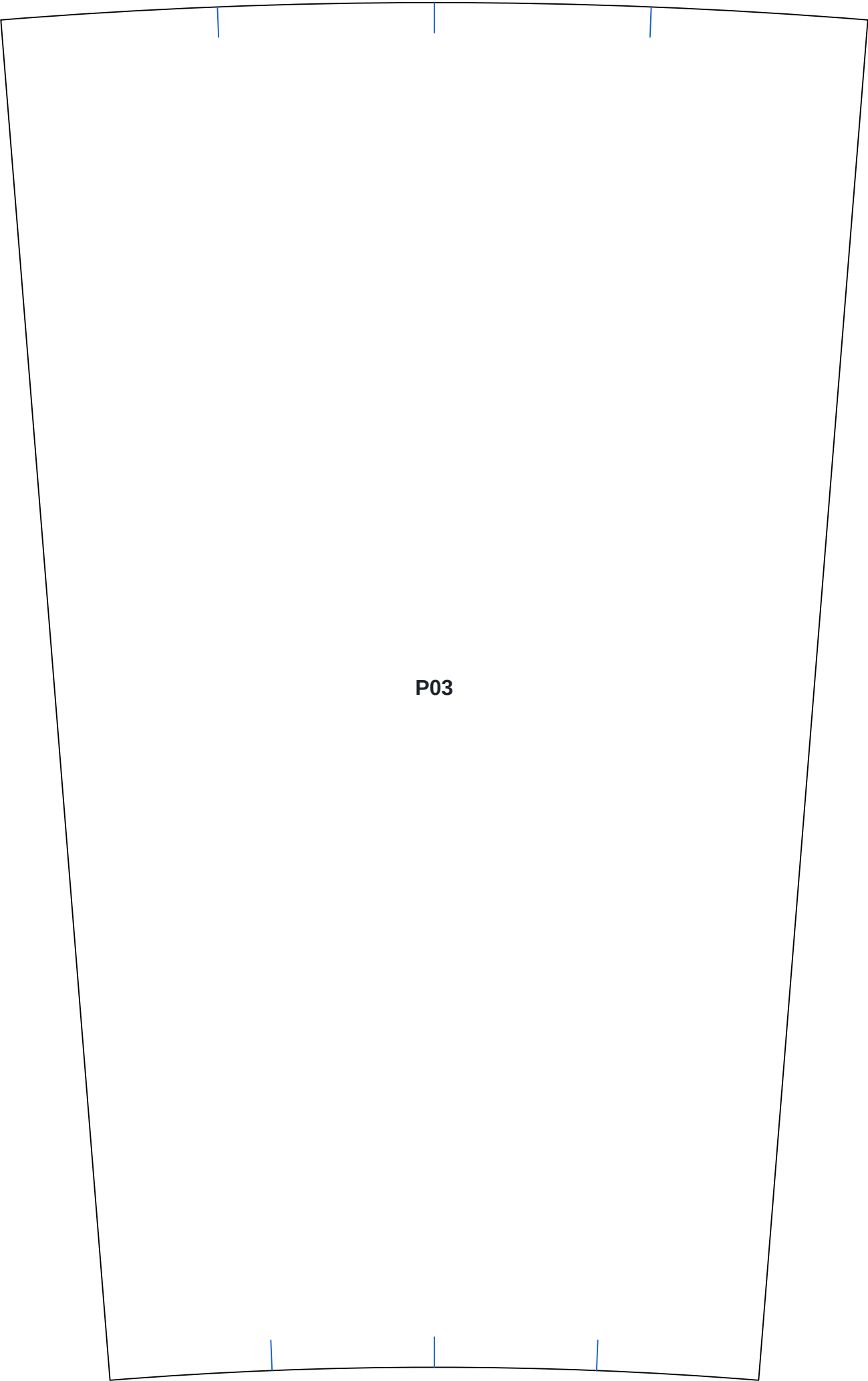
P02 Header (2 of 2)

Qty 1 · cone Ø40.1 to Ø40.9 mean, 144.4 long · outer R7078.26, inner R6933.85, 1.040° sector, slant 144.41
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14



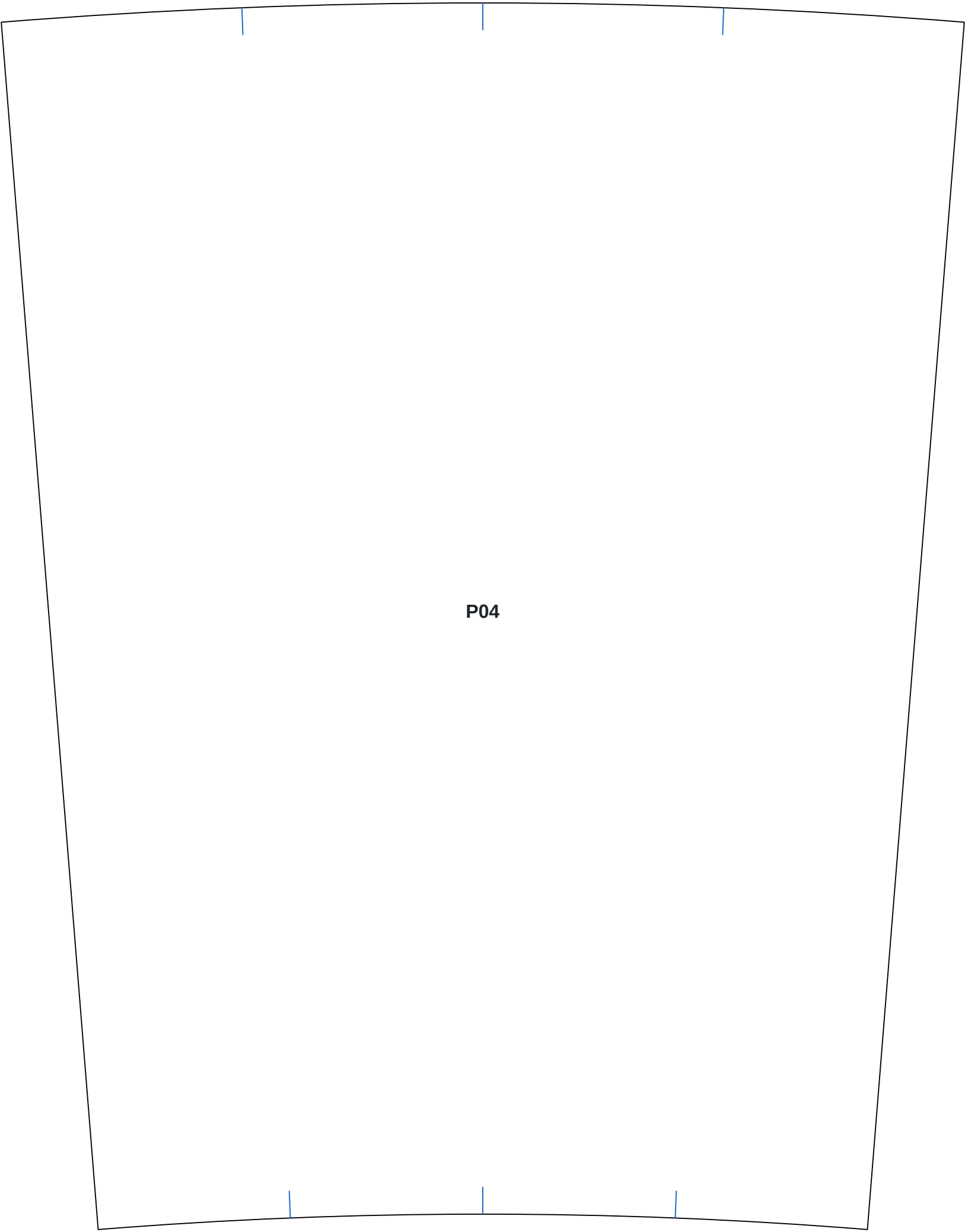
P03 Diffuser 1 (1 of 2)

Qty 1 · cone Ø40.9 to Ø54.7 mean, 270.1 long · outer R1072.61, inner R802.42, 9.175° sector, slant 270.19
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14



P04 Diffuser 1 (2 of 2)

Qty 1 · cone Ø54.7 to Ø68.4 mean, 270.1 long · outer R1342.80, inner R1072.61, 9.175° sector, slant 270.19
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14

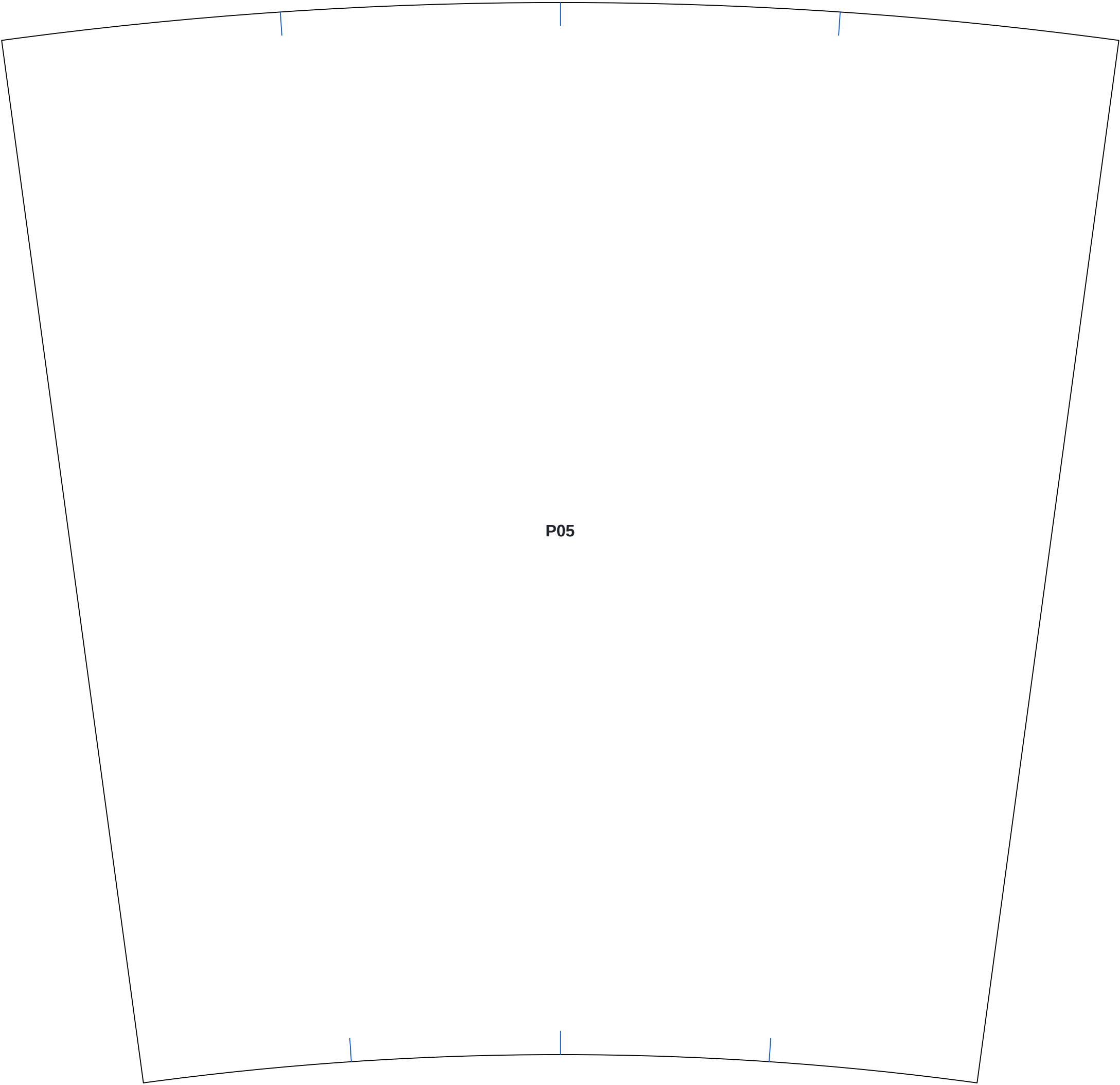


100 mm — print at 100% (actual size) and check this bar

P05 Diffuser 2 (1 of 2)

Qty 1 · cone Ø68.4 to Ø91.7 mean, 270.1 long · outer R1065.24, inner R794.89, 15.499° sector, slant 270.35

JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14

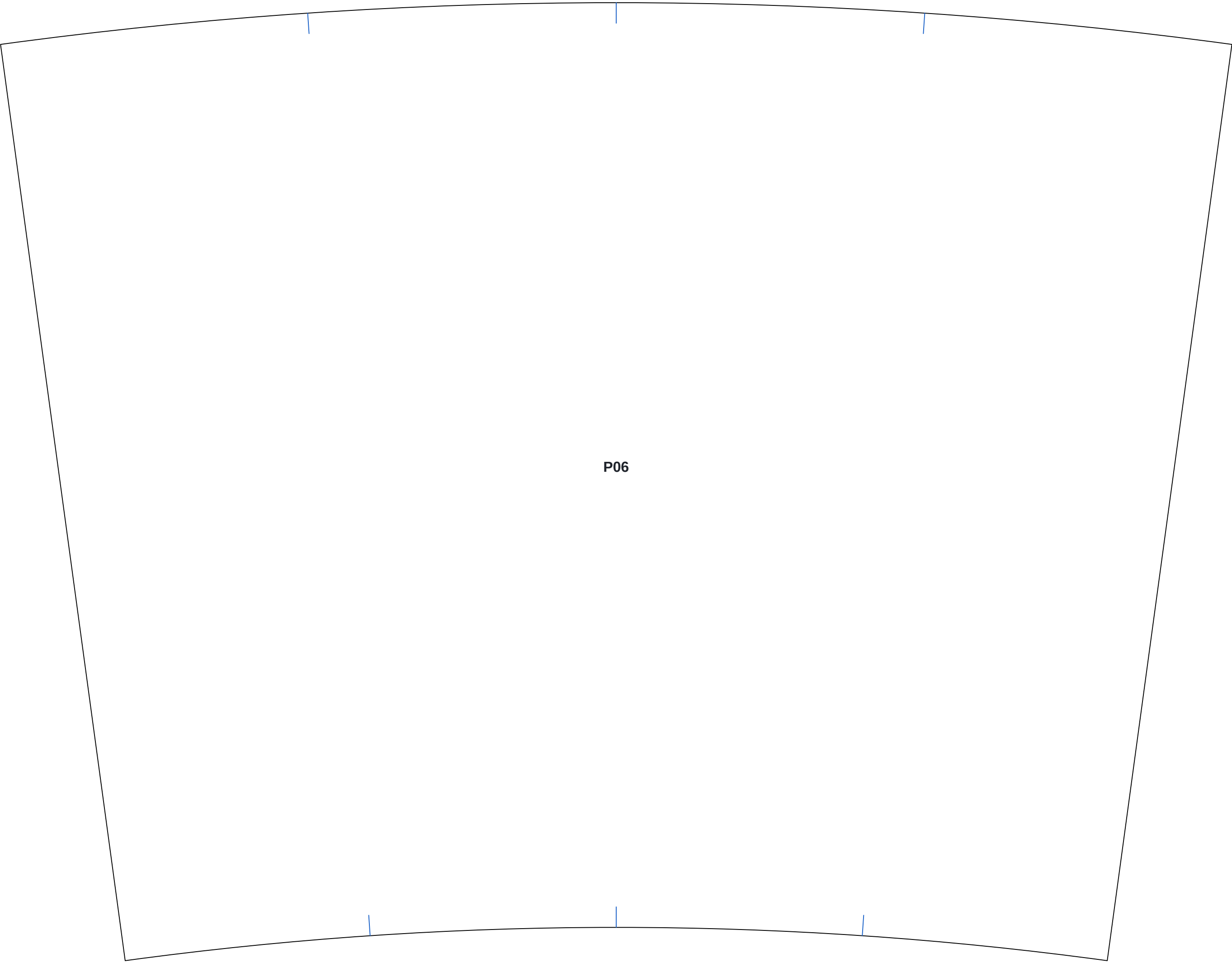


100 mm — print at 100% (actual size) and check this bar

P06 Diffuser 2 (2 of 2)

Qty 1 · cone Ø91.7 to Ø115.0 mean, 270.1 long · outer R1335.59, inner R1065.24, 15.499° sector, slant 270.35

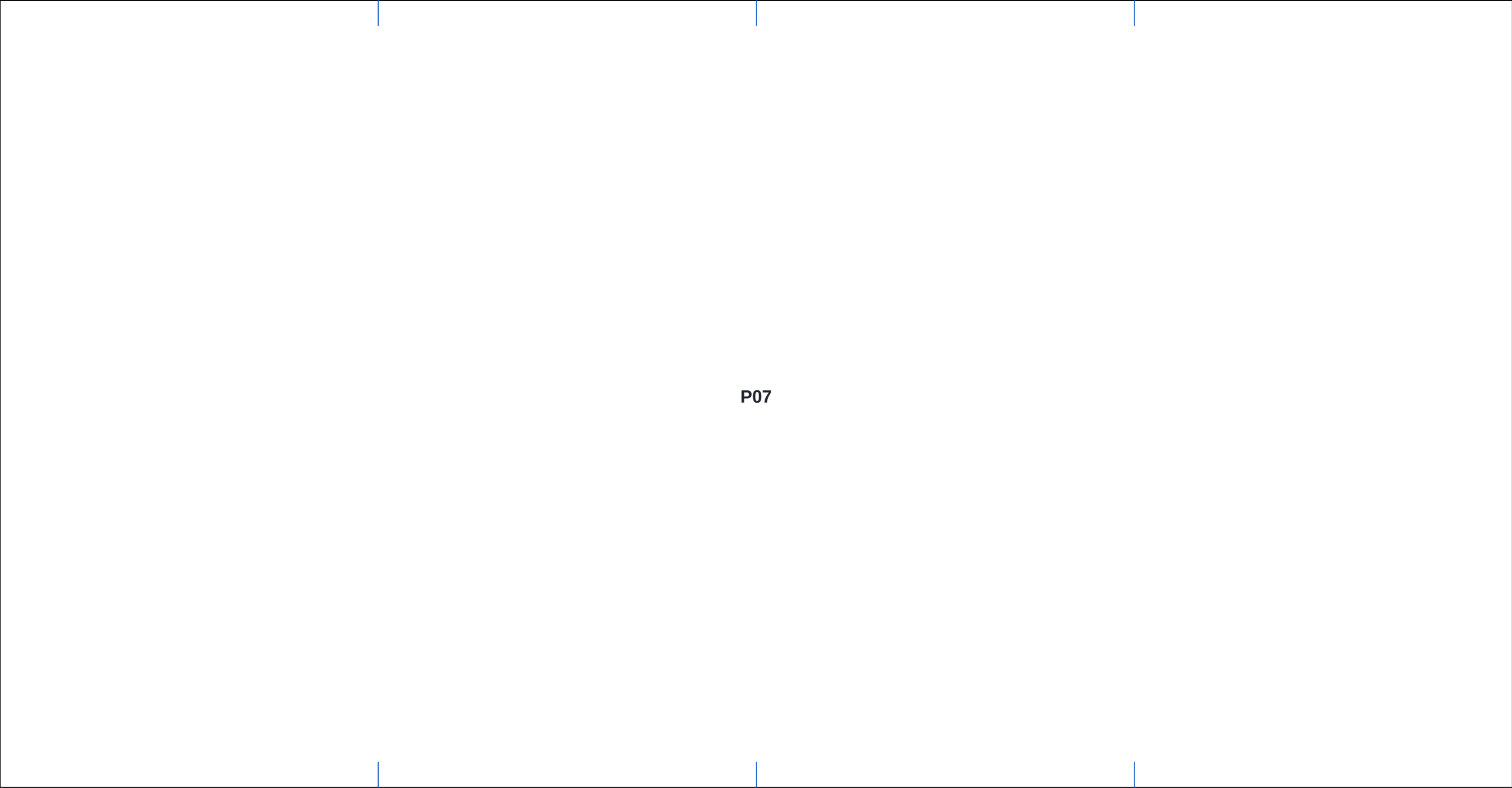
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14



100 mm — print at 100% (actual size) and check this bar

P07 Belly (1 of 2)

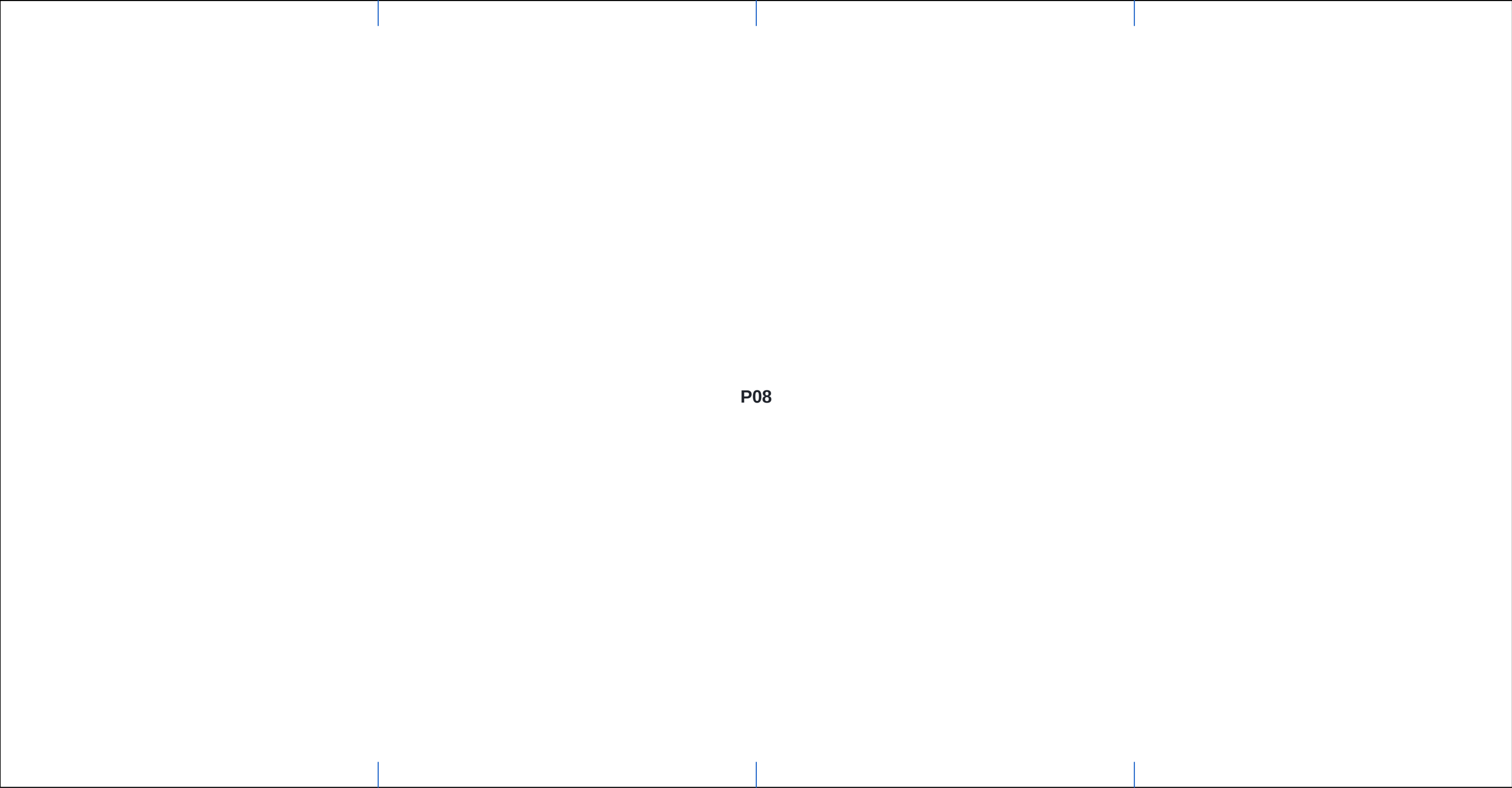
Qty 1 · cylinder Ø115.0 mean, 187.9 long · flat 361.28 × 187.90
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14



100 mm — print at 100% (actual size) and check this bar

P08 Belly (2 of 2)

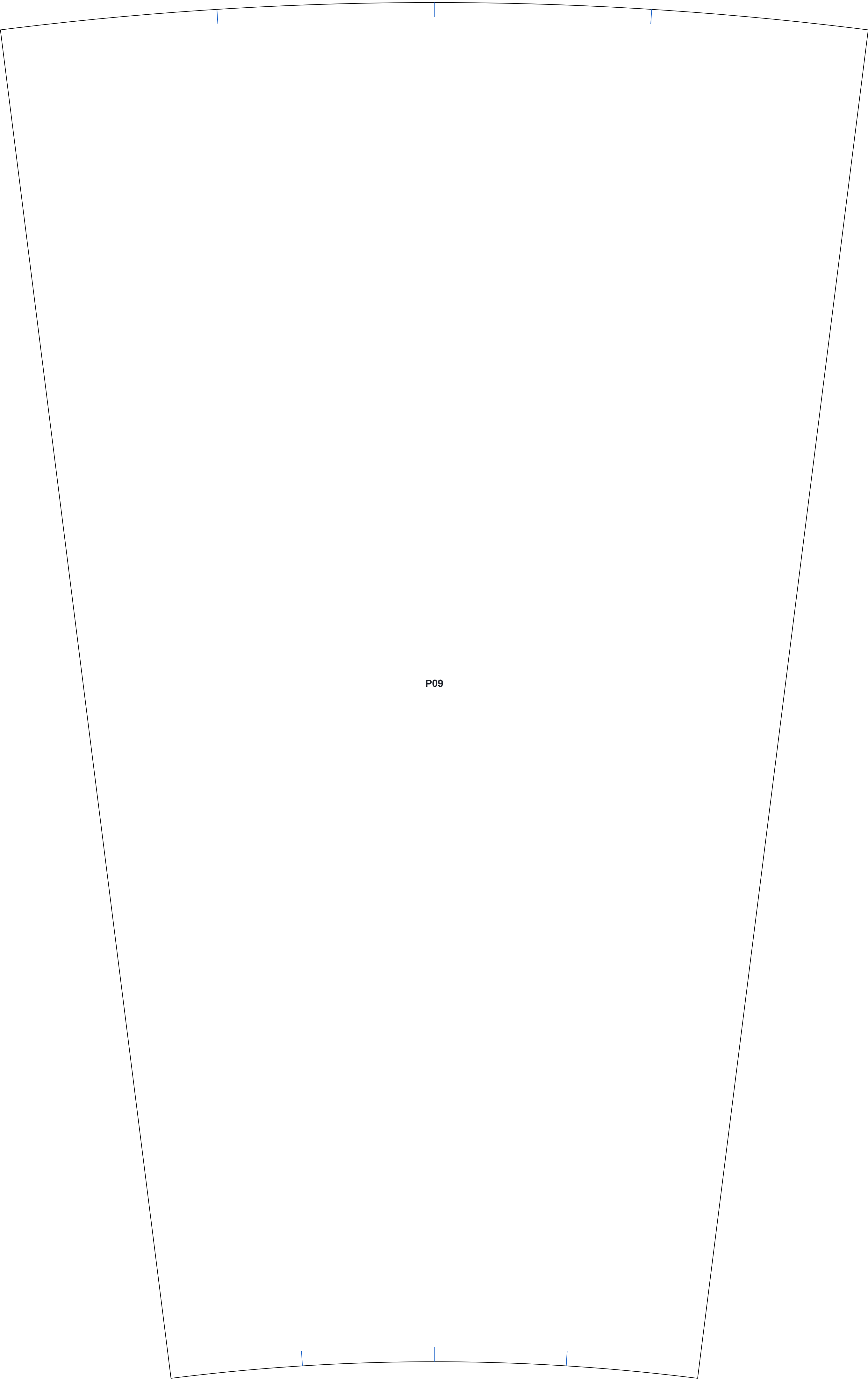
Qty 1 · cylinder Ø115.0 mean, 187.9 long · flat 361.28 × 187.90
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14



100 mm — print at 100% (actual size) and check this bar

P09 Baffle (1 of 2)

Qty 1 · cone Ø69.8 to Ø115.0 mean, 563.7 long · outer R1434.68, inner R870.54, 14.428° sector, slant 564.14
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14



P10 Baffle (2 of 2)

Qty 1 · cone Ø24.6 to Ø69.8 mean, 563.7 long · outer R870.54, inner R306.40, 14.428° sector, slant 564.14
JLO L372 generator chamber · 1.0 mm Mild steel · pipegen 2026.09.14

